

SMART PARKING SYSTEM

Occupancy Detection using Arduino

Group 3

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1 Abstract

The Smart Parking System is an Arduino-based solution designed to automate parking management using real-time vehicle detection. It uses ultrasonic sensors, a servo motor, LEDs, and a buzzer to control entry and exit while maintaining a vehicle count. This system reduces congestion and improves efficiency.

2 Introduction

Finding parking in urban areas is a major challenge, leading to time loss and increased pollution. This project provides an automated system that detects vehicles and manages parking availability efficiently.

3 Problem Statement

Drivers spend significant time searching for parking, causing:

- Traffic congestion
- Fuel wastage
- Increased carbon emissions

4 Proposed Solution

The system provides an automated, sensor-based solution that:

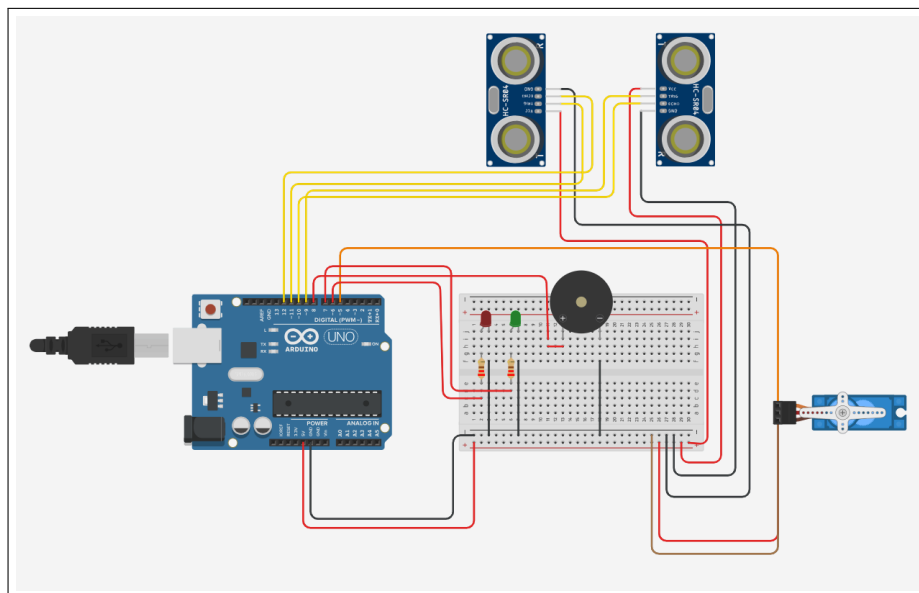
- Detects vehicles using ultrasonic sensors
- Controls gate using a servo motor
- Maintains real-time parking count
- Uses LEDs and buzzer for status indication

5 System Overview

5.1 Components

- Arduino UNO
- Ultrasonic Sensors (HC-SR04)
- Servo Motor
- LEDs and Buzzer

5.2 Circuit Diagram



6 Working Principle

- Entry sensor increases count and opens gate

- Exit sensor decreases count
- Green LED indicates availability
- Red LED and buzzer indicate full parking

7 Social Impact

7.1 Environmental Benefits

Reduces fuel consumption and CO₂ emissions by minimizing idle time.

7.2 Time Saving

Saves 15–20 minutes per trip, improving productivity and quality of life.

8 Applications

- Malls
- Airports
- Apartments
- Offices

9 Future Enhancements

- IoT-based monitoring
- Mobile application
- Display for slot count

10 Conclusion

The Smart Parking System provides an efficient and automated way to manage parking. It improves user experience and reduces environmental impact.